

STA141A Final Project Report

Team 5

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1. Introduction

Global carbon dioxide emissions increased significantly in 2018, raising concerns about their impact on climate change and the effectiveness of existing environmental policies. Recently, it has become clear how much of an impact this has on many different aspects of our lives.

The research question we would like to explore further examines how socioeconomic factors such as GDP and population impact CO2 emissions among countries. By exploring this relationship, we can assist the government in improving urban planning, resource allocation, and general social development.

By examining this relationship, our project aims to build a model that can predict CO2 emissions based on specific and significant factors.

2. Data Summary

The data set contains carbon dioxide emissions and related economic indicators for each country in 2018. We chose carbon dioxide emission (in million tons) as the dependent variable, and there are three explanatory variables in the dataset, including population, gross domestic product (in US \$), and oil carbon dioxide emissions (in million tons). It contains complete data points and variables with economic significance, which makes it easier to conduct analysis and models with reduced potential bias.

3. Exploring and Visualizing Data

3.1 Summary of Statistics

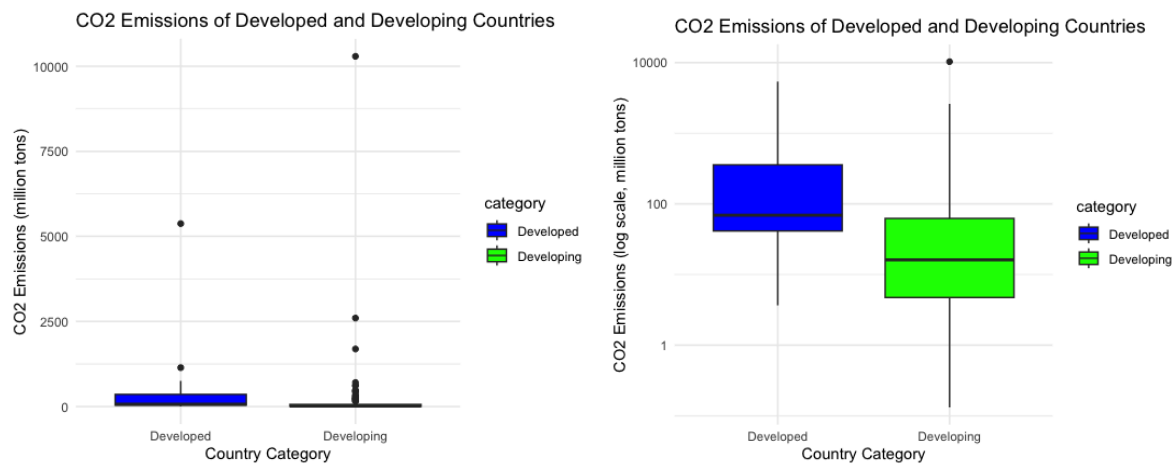
The table provides summary statistics for several variables in a dataset of 165 countries for the year 2018. It shows that CO2 emissions range from 0.132 to 10289.990 million tons, with a mean of 214.005 million tons and a median of 22.609 million tons. Population sizes vary widely, from 71,630 to 1.428 billion, with a mean of 45.92 million. GDP figures range from \$667.8 million to \$18.15 trillion, with a mean of \$690.3 billion and a median of \$100.4 billion. CO2 emissions from oil range from 0.132 to 2316.811 million tons, with a mean of 65.917 million tons and a median of 10.897 million tons. Iso_code and country are noted as categorical variables, so the summary does not provide the same statistics as other variables.

iso_code	country	year	co2	population	gdp	oil_co2
Length:165	Length:165	Min. :2018	Min. : 0.132	Min. :7.163e+04	Min. :6.678e+08	Min. : 0.132
Class :character	Class :character	1st Qu.:2018	1st Qu.: 6.371	1st Qu.:4.403e+06	1st Qu.:3.121e+10	1st Qu.: 3.420
Mode :character	Mode :character	Median :2018	Median : 22.609	Median :1.067e+07	Median :1.004e+11	Median : 10.897
NA	NA	Mean :2018	Mean : 214.005	Mean :4.592e+07	Mean :6.903e+11	Mean : 65.917
NA	NA	3rd Qu.:2018	3rd Qu.: 95.575	3rd Qu.:3.248e+07	3rd Qu.:4.313e+11	3rd Qu.: 34.105
NA	NA	Max. :2018	Max. :10289.990	Max. :1.428e+09	Max. :1.815e+13	Max. :2316.811

3.2 Boxplots of CO2 Emissions for Developed and Developing Countries

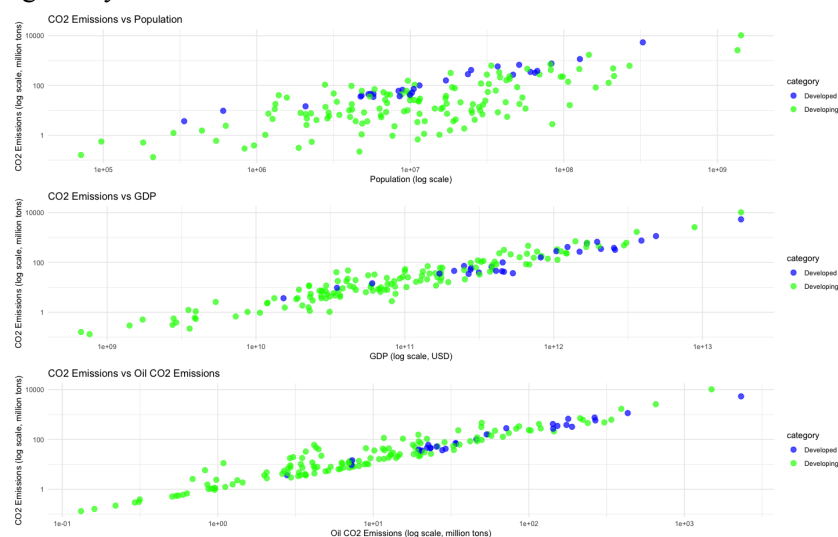
We categorized countries in the dataset into developed and developing countries based on the UN classification. 28 are noted as developed countries, and 137 are noted as developing countries. The two boxplots below show the distribution of CO2 emissions for developed and developing countries.

In the first boxplot (linear scale), it is challenging to compare the CO2 emissions because the values are highly skewed, with extreme outliers significantly influencing the visualization. Therefore, we decided to use a logarithmic scale to fix this issue. The second boxplot (logarithmic scale) provides a clearer comparison. Here, the median CO2 emissions for developed countries are slightly higher than those for developing countries. The spread (interquartile range) of CO2 emissions is broader for developed countries, indicating more variability. Both groups show significant outliers, but the logarithmic scale effectively compresses the range, making the distribution more interpretable. This boxplot indicates that developed countries tend to emit more CO2 gasses than developing countries.



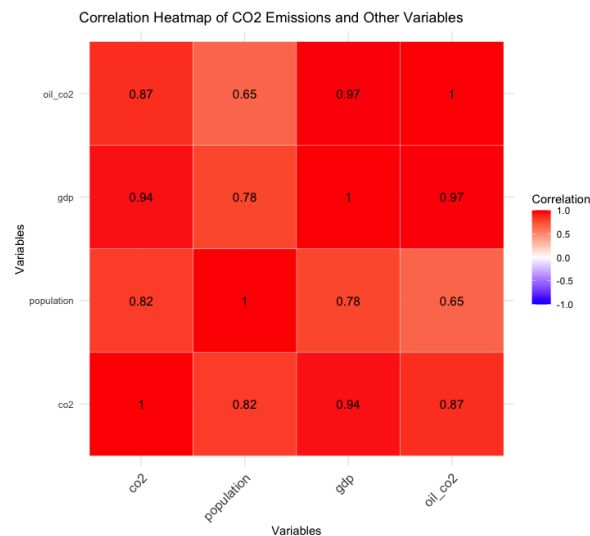
3.3 Scatterplots

The scatter plots reveal positive correlations between CO2 emissions and population, GDP, and oil CO2 emissions, with a logarithmic scale for CO2 emissions enhancing visibility due to the same concern as in 3.2. Developed countries (blue) generally exhibit higher CO2 emissions compared to developing countries (green) at similar levels of population and GDP. The strong correlation between oil CO2 emissions and total CO2 emissions highlights oil usage as a significant contributor to overall emissions in both groups. These relationships indicate that economic activity and energy consumption are key drivers of CO2 emissions globally.



3.4 Correlation Heatmap Between Variables

From the correlation heatmap below, it shows strong positive correlations between CO2 emissions and other variables: GDP (0.94), population (0.82), and oil CO2 emissions (0.87). GDP and oil CO2 emissions are also highly correlated with each other (0.97). This indicates that higher economic activity (GDP), larger populations, and greater oil consumption are strongly associated with increased CO2 emissions. The high correlations highlight the interconnectedness of economic and energy factors in driving CO2 emissions. On the other hand, this brings the potential concerns about multicollinearity and interaction effects when performing regression analysis.



4. Model Selection

4.1 Linear Regression Analysis

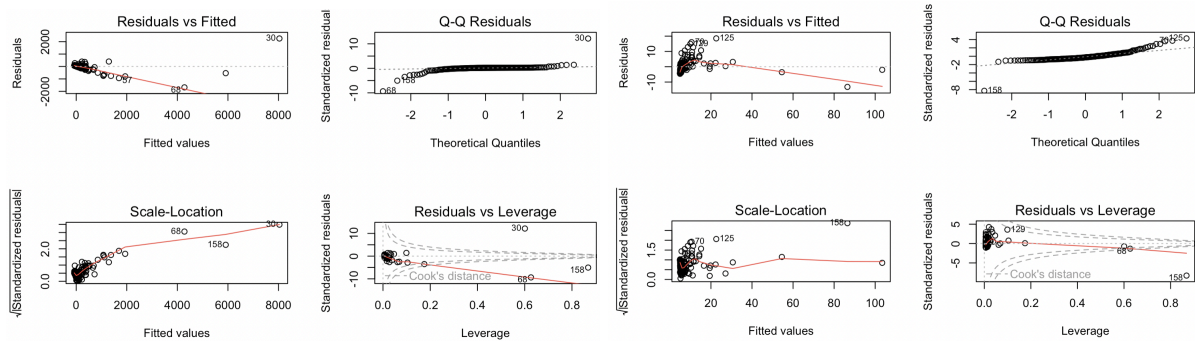
This section examines the relationship between CO2 emissions and various factors including population, oil CO2 emissions, and GDP. The goal is to develop a linear regression model that accurately predicts CO2 emissions based on these predictors. To improve the model's fit and meet the assumptions of linear regression, various transformations were applied to the response variable, CO2 emissions. These transformations included the square root, log, and Box-Cox transformations. The Box-Cox transformation was selected as it provided the best model fit. The Box-Cox transformation was applied to the response variable to stabilize variance and make the data more normal.

The intercept value is statistically significant, indicating a substantial baseline level of Box-Cox transformed CO2 emissions. The coefficient for population is not statistically significant, suggesting that population size does not have a significant impact on CO2 emissions in this model. The coefficient for oil CO2 emissions is not statistically significant, suggesting that oil CO2 emissions are not a significant predictor of overall CO2 emissions. The coefficient for GDP is statistically significant, indicating that GDP is a significant predictor of CO2 emissions. A higher GDP is associated with increased CO2 emissions.

Table 4.1.1 Box-Cox Results

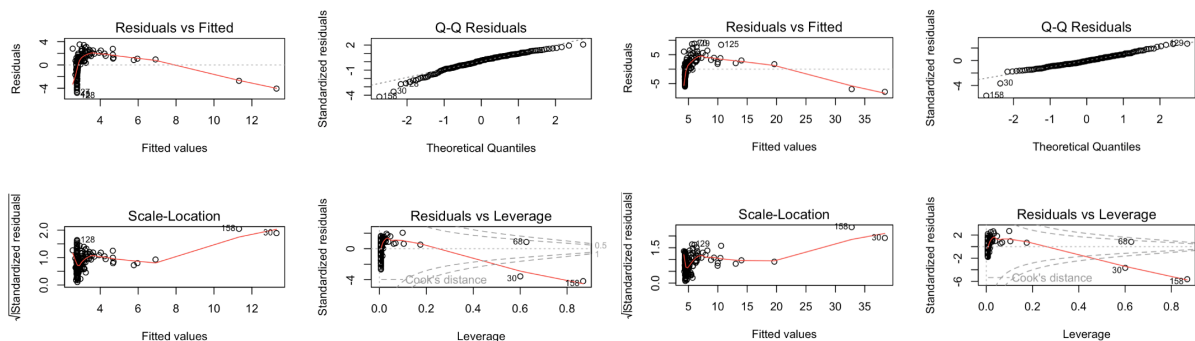
Coefficient	Estimate	Std. Error	t-value	p-value
Intercept	4.414e+00	2.774e-01	15.912	< 2e-16
population	-3.293e-09	4.034e-09	-0.816	0.415575
oil_co2	-1.124e-02	6.959e-03	-1.616	0.108134
gdp	3.059e-12	8.902e-13	3.436	0.000752

The Original Model issues with non-linearity, non-normality, and heteroscedasticity. Square Root Transformation improved linearity and heteroscedasticity. However there are some residual non-normality and influential points. Log Transformation has better linearity, normality, and heteroscedasticity than the original and square root models. It also makes some improvements in influential points. Box-Cox Transformation has significant improvement in all diagnostic plots. It Better handles linearity, normality, and heteroscedasticity.



Plot 4.1.1 Original Model Diagnostic

Plot 4.1.2 Square Root Transformation



Plot 4.1.3 Log Transformation

Plot 4.1.4 Box-Cox Transformation

The Box-Cox transformation provided the best fit for the linear regression model, addressing issues of non-linearity, non-normality, and heteroscedasticity. The model indicates that GDP is a significant predictor of CO2 emissions, while population and oil CO2 emissions are not statistically significant predictors in this model.

4.2 Logistic Model

The objective of this analysis is to examine the relationship between CO2 emissions and country type (developed vs. developing) using a logistic regression model. We aim to determine whether being a developed or developing country significantly affects the likelihood of having high CO2 emissions. The dataset includes variables such as the country name, CO2 emissions, population, GDP, and CO2 emissions from oil. For this analysis, we focused on the following steps:

1. **Classification of Countries:** Countries were categorized as "Developed" or "Developing" based on a predefined list from recognized sources like the United Nations.
2. **Categorization of CO2 Emissions:** CO2 emissions were categorized into "High" and "Low" based on the median value of CO2 emissions.

Table 4.2.1 Contingency Table

	Low	High
Developed	3	25
Developing	80	57

We employed a logistic regression model to analyze the relationship between CO2 emissions and country type. The dependent variable is the CO2 emission category (High/Low), and the independent variable is the country type (Developed/Developing).

Table 4.2.2 Results

Coefficient	Estimate	Standard.Error	z.value	p.value
(Intercept)	2.1203	0.6110	3.470	0.000520
Country_TypeDeveloping	-2.4592	0.6351	-3.872	0.000108

The log-odds of a country having high CO2 emissions when it is developed is approximately 2.1203. This suggests that developed countries have a higher likelihood of high CO2 emissions. The coefficient is -2.4592, indicating that the log odds of a country with high CO2 emissions decrease by approximately 2.4592 when the country is developing compared to developed. This suggests that developing countries are significantly less likely to have high CO2 emissions compared to developed countries. Both coefficients are highly significant (p-value < 0.001), providing strong evidence that country type (developed vs. developing) significantly affects CO2 emissions.

5. Model Validation

The result of the LDA shows the prior probabilities of the groups (this is before using any of the predictor variables) and the group means. The group means for the “low” value represents instances in which the CO2 emissions are below a specific threshold while it is the opposite for the “high” value.

5.1 Full Model

For the full model, we used all three predictor variables: population, GDP, and oil CO2, to understand the relationships between all of them and the response variable and get a deeper understanding of how they influence CO2 emissions.

5.1.1 Confusion Matrices

LOOCV confusion matrix values:

True Positives (TP): 3 False Negatives (FN): 74 False Positives (FP): 80

True Negatives (TN): 8 Accuracy Rate: 0.9333333

This indicates that 93.33% of the model predictions are accurate.

K-fold (10) Confusion Matrix values:

True Positives (TP): 4 False Negatives (FN): 74 False Positives (FP): 79

True Negatives (TN): 8 Accuracy Rate: 0.9272059

This indicates that 92.72% of the model predictions are accurate.

K-fold (20) Confusion Matrix:

True Positives (TP): 4 False Negatives (FN): 75 False Positives (FP): 79

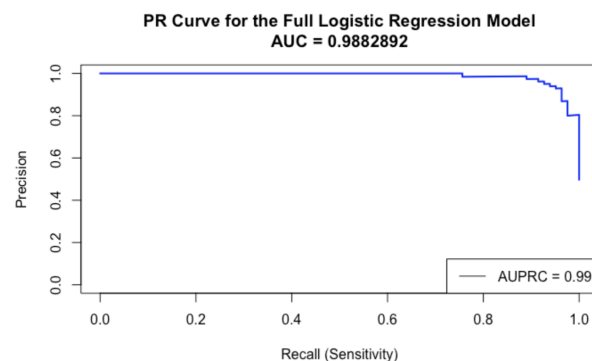
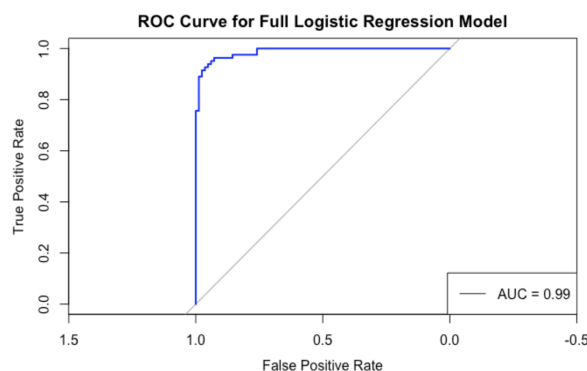
True Negatives (TN): 7 Accuracy Rate: 0.9277778

This indicates that 92.77% of the model predictions are accurate.

Doing the Leave-One-Out cross-validation and K-fold cross-validation for 2 splits and getting similar results for the accuracy rates shows that our full model is consistent with its predictions.

5.1.2. ROC Curve and PR Curve

We plotted the ROC curves for both the full and reduced model to see their performance as well as to compare them with each other. The AUC (Area Under The Curve) is 0.99. Since the closer the value is to 1, the better the model is, this shows that our full logistic regression model is extremely accurate and reliable.



5.2. Reduced Model

Making the reduced model was similar, except that we made it simpler by removing one of the predictor variables (oil CO₂), as we felt that the other two predictors were significantly more important.

5.2.1. Confusion Matrices

LOOCV confusion matrix values:

True Positives (TP): 2 False Negatives (FN): 73 False Positives (FP): 81

True Negatives (TN): 9 Accuracy Rate: 0.9333333

This indicates that 93.33% of the model predictions are accurate.

K-fold (10) Confusion Matrix values:

True Positives (TP): 2 False Negatives (FN): 72 False Positives (FP): 81

True Negatives (TN): 10 Accuracy Rate: 0.9272059

This indicates that 92.72% of the model predictions are accurate.

K-fold (20) Confusion Matrix values:

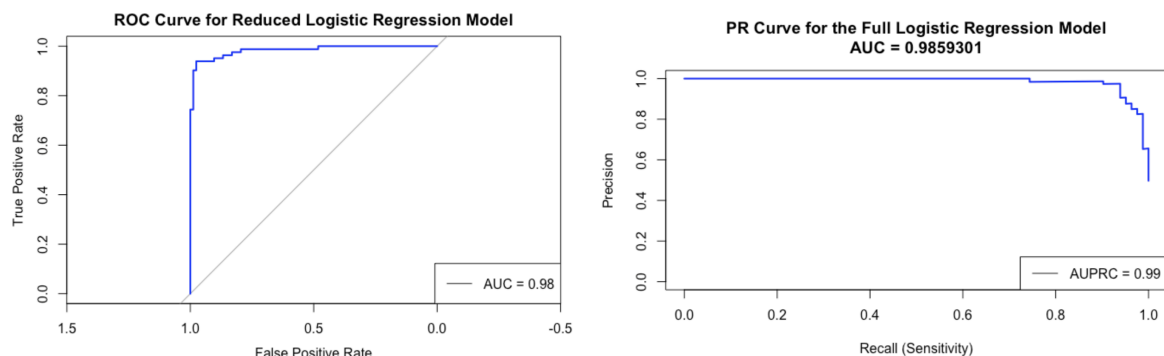
True Positives (TP): 2 False Negatives (FN): 73 False Positives (FP): 81

True Negatives (TN): 9 Accuracy Rate: 0.9333333

This indicates that 93.33% of the model predictions are accurate.

Similar to the full model, doing the Leave-One-Out cross-validation and K-fold cross-validation for 2 splits and getting similar results for the accuracy rates shows that our reduced model is consistent with its predictions. Both models have similar rates of accuracy.

5.2.2 ROC Curve and PR Curve



The AUC value of 0.98, which is very close to 1, shows the accuracy of the reduced model.

From the curves and confusion matrices, we can see that both models have extremely similar and high accuracy rates.

6. Conclusion

Within our analysis of the dataset containing carbon dioxide emissions and economic indicators for 165 countries in 2018, we found three key findings that could assist in mitigating the impact of climate change.

In terms of socioeconomic factors, we found that the correlation analysis displayed strong positive relationships between carbon dioxide emissions and GDP, oil carbon dioxide emissions, and population. It is shown that a higher GDP and larger population sizes are associated with increased carbon emissions. This may imply that population size and economic activity both play a massive role in driving carbon emissions towards an upward trend.

Another key finding that we found was through our boxplots. We categorized the countries into developed and developing countries and discovered that the developed countries often displayed higher levels of carbon emissions compared to the developing countries. The boxplots on a logarithmic scale embody the idea that developed nations are more likely to emit carbon emissions because of their energy consumption patterns as well as their industrial activities.

Our last key finding was with the regression analysis. The logistic regression model reveals that the economic status of a country plays a vital role in its carbon emission levels. This not only supports the key findings from the boxplots but also emphasizes the significance of whether or not the country is developed.

Overall the findings from our analysis underscore the importance of tailored environmental policies. For the developed countries, the focus should be shifted towards improving energy efficiency as well as implementing cleaner technologies. On the other hand, developing countries may benefit from more industrial activities, but finding a sustainable way to do so. Our findings highlight the complicated relationship between socioeconomic factors and carbon emissions which allows us to better address the global challenge of mitigating climate change.

Appendix

```
````{r}
data <- read.csv("~/Desktop/UCD_HW/STA141A/final project/CO2_2018.csv")
library(ggplot2)
library(readr)
library(knitr)
library(gridExtra)
library(reshape2)
library(tidyverse)
library(MASS)
library(dplyr)
library(pROC)
library(PRROC)
library(e1071)
sapply(data, function(x)
sum(is.na(x)))
````

````{r}
EXPLORING AND VISUALIZING DATA

Summary of Statistics
sum_data <- summary(data)
kable(sum_data)

Boxplots of developed and developing countries
developed_countries <- c("Australia", "Austria", "Belgium", "Canada", "Denmark", "Finland", "France",
"Germany",
 "Iceland", "Ireland", "Italy", "Japan", "Luxembourg", "Netherlands", "New Zealand",
"Norway",
 "Portugal", "Spain", "Sweden", "Switzerland", "United Kingdom", "United States",
 "Greece", "South Korea", "Israel", "Slovenia", "Singapore", "Taiwan")
data$country_category <- ifelse(data$country %in% developed_countries, "Developed", "Developing")
developed_data <- subset(data, country_category == "Developed")
developing_data <- subset(data, country_category == "Developing")

ggplot(data, aes(x = country_category, y = co2, fill = country_category)) +
 geom_boxplot() +
 labs(title = "CO2 Emissions of Developed and Developing Countries",
 x = "Country Category",
 y = "CO2 Emissions (million tons)") +
 theme_minimal() +
```

```

scale_fill_manual(values = c("Developed" = "blue", "Developing" = "green"))

ggplot(data, aes(x = country_category, y = co2, fill = country_category)) +
 geom_boxplot() +
 scale_y_log10() +
 labs(title = "CO2 Emissions of Developed and Developing Countries",
 x = "Country Category",
 y = "CO2 Emissions (log scale, million tons)") +
 theme_minimal() +
 scale_fill_manual(values = c("Developed" = "blue", "Developing" = "green"))

Scatterplots
plot1 <- ggplot(data, aes(x = population, y = co2, color = country_category)) +
 geom_point(alpha = 0.7, size = 3) +
 scale_x_log10() + scale_y_log10() +
 labs(title = "CO2 Emissions vs Population",
 x = "Population (log scale)",
 y = "CO2 Emissions (log scale, million tons)") +
 theme_minimal() +
 scale_color_manual(values = c("Developed" = "blue", "Developing" = "green"))

plot2 <- ggplot(data, aes(x = gdp, y = co2, color = country_category)) +
 geom_point(alpha = 0.7, size = 3) +
 scale_x_log10() + scale_y_log10() +
 labs(title = "CO2 Emissions vs GDP",
 x = "GDP (log scale, USD)",
 y = "CO2 Emissions (log scale, million tons)") +
 theme_minimal() +
 scale_color_manual(values = c("Developed" = "blue", "Developing" = "green"))

plot3 <- ggplot(data, aes(x = oil_co2, y = co2, color = country_category)) +
 geom_point(alpha = 0.7, size = 3) +
 scale_x_log10() + scale_y_log10() +
 labs(title = "CO2 Emissions vs Oil CO2 Emissions",
 x = "Oil CO2 Emissions (log scale, million tons)",
 y = "CO2 Emissions (log scale, million tons)") +
 theme_minimal() +
 scale_color_manual(values = c("Developed" = "blue", "Developing" = "green"))

grid.arrange(plot1, plot2, plot3, ncol = 1)

Correlation
selected_data <- data[, c("co2", "population", "gdp", "oil_co2")]
cor_matrix <- cor(selected_data, use = "complete.obs")

```

```

melted_cor_matrix <- melt(cor_matrix)
ggplot(data = melted_cor_matrix, aes(x = Var1, y = Var2, fill = value)) +
 geom_tile(color = "white") +
 scale_fill_gradient2(low = "blue", high = "red", mid = "white",
 midpoint = 0, limit = c(-1, 1), space = "Lab",
 name="Correlation") +
 geom_text(aes(label = round(value, 2)), color = "black", size = 4) + # Add correlation values
 theme_minimal() +
 theme(axis.text.x = element_text(angle = 45, vjust = 1,
 size = 12, hjust = 1)) +
 coord_fixed() +
 labs(title = "Correlation Heatmap of CO2 Emissions and Other Variables",
 x = "Variables",
 y = "Variables")
````

```

```

```` {r}
MODEL SELECTION

Correlation matrix
cor_matrix <- cor(data[, c("co2", "population", "gdp", "oil_co2")], use = "complete.obs")
print(cor_matrix)

Fit the linear regression model
linear_model <- lm(co2 ~ population + oil_co2 + gdp, data = data)

Summary of the model
summary(linear_model)

Diagnostic plots
par(mfrow = c(2, 2))
plot(linear_model)

Apply square root transformation to the response variable
data$sqrt_co2 <- sqrt(data$co2)

Fit the linear regression model with the transformed response variable
linear_model_sqrt <- lm(sqrt_co2 ~ population + oil_co2 + gdp, data = data)

Summary of the model
summary(linear_model_sqrt)

Diagnostic plots
par(mfrow = c(2, 2))

```

```

plot(linear_model_sqrt)

Apply log transformation to the response variable
data$log_co2 <- log(data$co2)

Fit the linear regression model with the transformed response variable
linear_model_log <- lm(log_co2 ~ population + oil_co2 + gdp, data = data)

Summary of the model
summary(linear_model_log)

Diagnostic plots
par(mfrow = c(2, 2))
plot(linear_model_log)

Apply Box-Cox transformation to find the optimal lambda
boxcox_model <- boxcox(lm(co2 ~ population + oil_co2 + gdp, data = data))
optimal_lambda <- boxcox_model$x[which.max(boxcox_model$y)]

Transform the response variable using the optimal lambda
data$boxcox_co2 <- (data$co2^optimal_lambda - 1) / optimal_lambda

Fit the linear regression model with the transformed response variable
linear_model_boxcox <- lm(boxcox_co2 ~ population + oil_co2 + gdp, data = data)

Summary of the model
summary(linear_model_boxcox)

Diagnostic plots
par(mfrow = c(2, 2))
plot(linear_model_boxcox)

Box-Cox transformation
boxcox_model <- boxcox(lm(co2 ~ population + oil_co2 + gdp, data = data))
optimal_lambda <- boxcox_model$x[which.max(boxcox_model$y)]

Transform the response variable using the optimal lambda
data$boxcox_co2 <- (data$co2^optimal_lambda - 1) / optimal_lambda

Fit the linear regression model with the transformed response variable
linear_model_boxcox <- lm(boxcox_co2 ~ population + oil_co2 + gdp, data = data)

Summary of the model
summary(linear_model_boxcox)

```

```

Diagnostic plots
par(mfrow = c(2, 2))
plot(linear_model_boxcox)

Define a threshold to categorize CO2 emissions
threshold <- median(data$co2, na.rm = TRUE)

Create a new binary variable for CO2 emissions
data <- data %>%
 mutate(CO2_Category = ifelse(co2 > threshold, "High", "Low"),
 CO2_Category = factor(CO2_Category, levels = c("Low", "High")))

Fit the logistic regression model
logistic_model <- glm(CO2_Category ~ country_category, data = data, family = binomial)

Summary of the model
summary(logistic_model)

Create the contingency table data
contingency_data <- matrix(c(3, 25, 80, 57), nrow = 2, byrow = TRUE)
rownames(contingency_data) <- c("Developed", "Developing")
colnames(contingency_data) <- c("Low", "High")

Convert to data frame for better visualization
contingency_df <- as.data.frame(contingency_data)

Create the results table data
results_data <- data.frame(
 Coefficient = c("(Intercept)", "Country_TypeDeveloping"),
 Estimate = c(2.1203, -2.4592),
 `Standard Error` = c(0.6110, 0.6351),
 `z-value` = c(3.470, -3.872),
 `p-value` = c(0.000520, 0.000108)
)

Draw the results table
results_table <- tableGrob(results_data, theme = ttheme_default(base_size = 12))

Plot both tables side by side
grid.arrange(results_table, nrow = 1, top = "Logistic Regression Results")

```

```

```{r}
## MODEL VALIDATION ##

#LDA model
lda_model <- lda(CO2_Category ~ country_category, data = data)
print(lda_model)

#Full Model Confusion Matrix
co2_validation <- function(data, num_folds) {
  set.seed(100)
  folds <- cut(seq(1, nrow(data)), breaks = num_folds, labels = FALSE)
  accuracy <- numeric(num_folds)
  all_pred <- NULL
  all_actuals <- NULL

  for(i in 1:num_folds) {
    #Split data into training and testing sets
    testIndexes <- which(folds == i, arr.ind = TRUE)
    testData <- data[testIndexes, ]
    trainData <- data[-testIndexes, ]
    log_model <- glm(CO2_Category ~ population + gdp + oil_co2, data = trainData, family = "binomial")

    predictions <- predict(log_model, newdata = testData, type = "response")
    predictions_binary <- ifelse(predictions > 0.5, "High", "Low")
    accuracy[i] <- mean(predictions_binary == testData$CO2_Category)

    all_pred <- c(all_pred, predictions_binary)
    all_actuals <- c(all_actuals, testData$CO2_Category)
  }

  aggregated_confusion_matrix <- table(Predicted = all_pred, Actual = all_actuals)
  return(list(AverageAccuracy = mean(accuracy), ConfusionMatrix = aggregated_confusion_matrix))
}

# 3 types of cv: LOOCV, 10-fold, 20-fold
co2_loocv <- co2_validation(data, nrow(data))
co2_k10 <- co2_validation(data, 10)
co2_k20 <- co2_validation(data, 20)

co2_loocv$AverageAccuracy
co2_loocv$ConfusionMatrix
co2_k10$AverageAccuracy
co2_k10$ConfusionMatrix
co2_k20$AverageAccuracy

```

```
co2_k20$ConfusionMatrix
```

```
#Full model ROC curve
```

```
full_model <- glm(CO2_Category ~ population + gdp + oil_co2, data = data, family = "binomial")
```

```
predicted_probabilities_full <- predict(full_model, type = "response")
```

```
roc_curve_full <- roc(ifelse(data$CO2_Category == "High", 1, 0), predicted_probabilities_full)
```

```
plot(roc_curve_full, main = "ROC Curve for Full Logistic Regression Model",
```

```
      xlab = "False Positive Rate", ylab = "True Positive Rate",
```

```
      col = "blue", lwd = 2)
```

```
# AUC value
```

```
auc_value_full <- round(auc(roc_curve_full), 2)
```

```
legend("bottomright", legend = paste("AUC =", auc_value_full), col = "black", lwd = 1)
```

```
#Reduced Model Confusion Matrix
```

```
co2_validation <- function(data, num_folds) {
```

```
  set.seed(100)
```

```
  folds <- cut(seq(1, nrow(data)), breaks = num_folds, labels = FALSE)
```

```
  accuracy <- numeric(num_folds)
```

```
  all_pred <- NULL
```

```
  all_actuals <- NULL
```

```
  for(i in 1:num_folds) {
```

```
    #Split data into training and testing sets
```

```
    testIndexes <- which(folds == i, arr.ind = TRUE)
```

```
    testData <- data[testIndexes, ]
```

```
    trainData <- data[-testIndexes, ]
```

```
    #no oil_co2 for reduced model
```

```
    log_model <- glm(CO2_Category ~ population + gdp, data = trainData, family = "binomial")
```

```
    predictions <- predict(log_model, newdata = testData, type = "response")
```

```
    predictions_binary <- ifelse(predictions > 0.5, "High", "Low")
```

```
    accuracy[i] <- mean(predictions_binary == testData$CO2_Category)
```

```
  all_pred <- c(all_pred, predictions_binary)
```

```
  all_actuals <- c(all_actuals, testData$CO2_Category)
```

```
}
```



```

aggregated_confusion_matrix <- table(Predicted = all_pred, Actual = all_actuals)

return(list(AverageAccuracy = mean(accuracy), ConfusionMatrix = aggregated_confusion_matrix))
}

# 3 types of cv: LOOCV, 10-fold, 20-fold
co2_loocv <- co2_validation(data, nrow(data))
co2_k10 <- co2_validation(data, 10)
co2_k20 <- co2_validation(data, 20)

co2_loocv$AverageAccuracy
co2_loocv$ConfusionMatrix
co2_k10$AverageAccuracy
co2_k10$ConfusionMatrix
co2_k20$AverageAccuracy
co2_k20$ConfusionMatrix

#Reduced model ROC curve
reduced_model <- glm(CO2_Category ~ population + gdp, data = data, family = "binomial")
predicted_probabilities_reduced <- predict(reduced_model, type = "response")

roc_curve_reduced <- roc(ifelse(data$CO2_Category == "High", 1, 0), predicted_probabilities_reduced)

plot(roc_curve_reduced, main = "ROC Curve for Reduced Logistic Regression Model",
     xlab = "False Positive Rate", ylab = "True Positive Rate",
     col = "blue", lwd = 2)

auc_value_reduced <- round(auc(roc_curve_reduced), 2)
legend("bottomright", legend = paste("AUC =", auc_value_reduced), col = "black", lwd = 1)

#Full model PR Curve
data <- data
data$CO2_Category <- factor(data$CO2_Category, levels = c("Low", "High"))
log_model <- glm(CO2_Category ~ population + gdp + oil_co2, data = data, family = "binomial")

# Predict probabilities
predicted_probabilities <- predict(log_model, newdata = data, type = "response")

# Plot PR curve
pr_curve <- pr.curve(scores.class0 = predicted_probabilities, weights.class0 = ifelse(data$CO2_Category
== "High", 1, 0), curve = TRUE)
plot(pr_curve, main = "PR Curve for the Full Logistic Regression Model",
     xlab = "Recall (Sensitivity)", ylab = "Precision",
     col = "blue", lwd = 2)

```

```

auprc_value <- round(pr_curve$auc.integral, 2)
legend("bottomright", legend = paste("AUPRC =", auprc_value), col = "black", lwd = 1)

#Reduced model PR Curve
data <- data
data$CO2_Category <- factor(data$CO2_Category, levels = c("Low", "High"))
log_model <- glm(CO2_Category ~ population + gdp, data = data, family = "binomial")

# Predict probabilities
predicted_probabilities <- predict(log_model, newdata = data, type = "response")

# Plot PR curve
pr_curve <- pr.curve(scores.class0 = predicted_probabilities, weights.class0 = ifelse(data$CO2_Category
== "High", 1, 0), curve = TRUE)
plot(pr_curve, main = "PR Curve for the Full Logistic Regression Model",
      xlab = "Recall (Sensitivity)", ylab = "Precision",
      col = "blue", lwd = 2)
auprc_value <- round(pr_curve$auc.integral, 2)
legend("bottomright", legend = paste("AUPRC =", auprc_value), col = "black", lwd = 1)

```